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MACSE698 | DISSERTATION-1 | FINAL REVIEW

Tail-Sensitive Leakage Assessment for ML-KEM

Combining Extreme Value Theory with Deep Learning-Based Profiled Attacks

Presented by
G Madhusudan
Reg. No. 25MCY0001

Under the guidance of
Dr. Jeyanthi N
Professor Grade 1, SCORE

M.Tech Cyber Security | Department of Cyber Security and Intelligent Systems | School of Computer Science Engineering and Information Systems, VIT Vellore | 28 October 2026

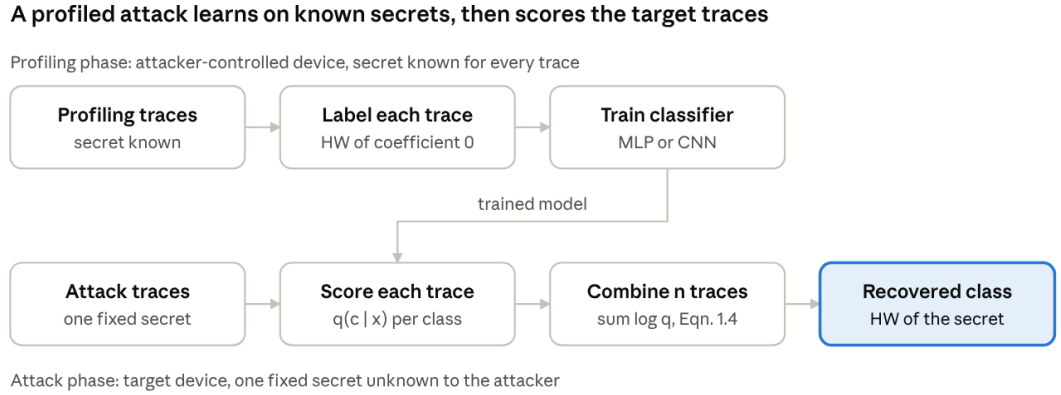
ML-KEM is secure on paper, but devices leak

ML-KEM (FIPS 203, Aug 2024) is the NIST post-quantum key-encapsulation standard derived from CRYSTALS-Kyber; its security rests on Module-LWE.

Implementation leakage: power and EM emanations depend on the data processed, so secrets can be recovered without breaking the mathematics.

Profiled attacks learn the leakage on a device with known secrets, then attack the victim's traces; deep networks dominate this setting (Picek et al., 2023).

This work: EM leakage of three ML-KEM implementations (Ledger Donjon dataset) assessed with leakage statistics, extreme value theory and deep-learning profiled attacks.



Profiled side-channel attack workflow (report Fig. 1.1)

Central idea

Security hinges on extremes (one leaking sample, one misleading trace), yet standard evaluation tools are built around averages.

Motivation: three weaknesses in current practice



Fixed detection thresholds

TVLA's |t| > 4.5 is applied independently at thousands of correlated samples; on this dataset it raises false alarms under repeated checking (Feretzakis & Papaspyridis, 2026).



Arbitrary point counts

Attacks keep a fixed number of points of interest, such as the top 50. The base paper used fixed TVLA filtering on an unprotected target only (Chinbat et al., 2025).



Misleading evaluation

Non-uniform class priors make guessing entropy report false success (Rousselot et al., 2026); networks are over-confident; validation-only results hide cross-session failure.

Missing: a data-driven rule for how many points carry real leakage, and attack results checked on independent recordings.

Objectives and research questions

Objectives

1. Hamming-weight leakage assessment (t-test, SNR) on all three implementations
2. GPD tail cut-off for points of interest versus a fixed top-50 rule
3. CNN profiled attacks evaluated on fixed-key attack traces
4. Verify ground truth and trace-label pairing at every stage
5. Maximum-accuracy attack under a leakage-free protocol; find the minimum points
6. Evaluate on the complete attack sets and diagnose any failure

Research questions

RQ1 Where, and how strongly, do the three implementations leak the HW of secret coefficient 0?

RQ2 Does a GPD tail cut-off pick fewer points than top-50, and at what accuracy cost?

RQ3 How accurate are MLP and CNN attacks with K = 50, 100, 150 points on validation, test and attack data?

RQ4 When a model that does well on validation data fails on attack data, what is the cause?

Literature survey: 18 papers from 2023 onwards

01 Deep learning dominates

DL profiled attacks are the main profiling method and reach high accuracy on unprotected Kyber (Picek et al., 2023; Chinbat et al., 2025; Ravi et al., 2023).

02 Fixed point-selection rules

POI selection uses average-case statistics such as TVLA filtering; no attack paper models the tail of the leakage statistic.

03 Detection calibration is open

$|t| > 4.5$ misbehaves under repeated checking; SNR distribution derived (des Noes, 2024); trace-wide corrections ignore correlation.

04 Metrics under scrutiny

Non-uniform priors cause false GE success; uncalibrated networks are over-confident (Rousselot et al., 2026; Nouraniboosjin & Ganji, 2024).

05 EVT not used for EM leakage

Applied to timing side channels and permutation p-values (Baez et al., 2025; Peschel et al., 2026), not to EM leakage statistics.

06 Masked targets rarely studied

Few masked evaluations (Dubrova et al., 2023; Ganesh et al., 2024); second-order CPA breaks mkm4 in about 200 traces.

Base paper and the gaps this work closes

Aspect	Chinbat et al. (2025)	This work
Target	FPGA Kyber NTT, power traces	ML-KEM base multiplication, Cortex-M4, EM traces
Protection	Unprotected only	Two unprotected + one first-order masked
POI selection	TVLA, fixed rule	Top-50 vs GPD cut-off; training-only weighted SNR
Evaluation	Validation accuracy	Validation, test split, full attack sets, prior-aware
Best result	96.64% accuracy	ref 94.77% validation, 94.65% attack set

G1	**POI count fixed by rule**	GPD cut-off; K sweep
G2	**Validation-only results**	Test split + full attack sets
G3	**Misleading priors**	Baseline, null model, prior correction
G4	**Over-confident models**	Temperature scaling
G5	**Session shift undiagnosed**	Drift check, expected-HW diagnosis
G6	**Masked target rare**	mkm4 with share positive control
G7	**Ground truth unaudited**	Integrity audit, label fix

Dataset: Ledger Donjon ML-KEM EM traces

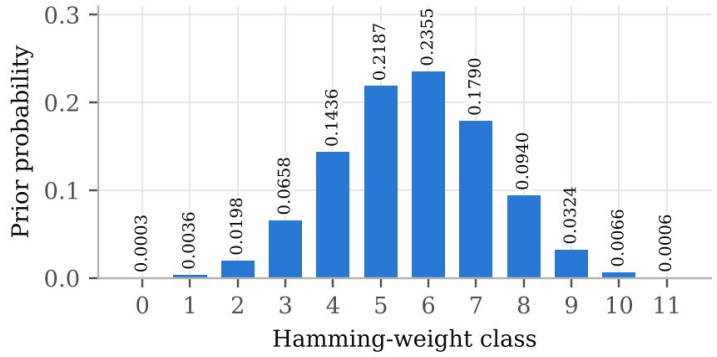
Implementation	Protection	Profiling	Attack	Samples	Fixed secret (HW)
pqm4	None	200,000	200,000	20,000	679 (HW 6)
ref	None	100,000	100,000	13,000	1026 (HW 2)
mkm4	1st-order masking	100,000	100,000	10,000 / share	1681 (HW 5)

800,000
traces analysed (all files)

12
Hamming-weight classes of coefficient 0

23.6%
majority baseline (HW 6 prior 0.2355)

Captured with a near-field EM probe on an STM32F407 (ARM Cortex-M4). Each trace covers one base multiplication r = a × b; the secret is operand a (pqm4, ref) or b split into two shares (mkm4).



HW class prior over values mod 3329 (report Fig. 1.2)

Why the baseline matters
Always guessing HW 6 is already 23.6% accurate, so every model is judged against this, not against 1/12 = 8.3%.

Techniques used in the pipeline

Leakage statistics

Welch's t-test between HW ≤ 6 and HW > 6 (TVLA level $|t| > 4.5$) and the class-weighted SNR across the 12 HW classes, streamed over full profiling sets.

Extreme value theory

Excesses of $|t|$ over u (98th percentile) fitted with a Generalised Pareto Distribution; each sample scored by $-\log_{10}$ of its survival probability.

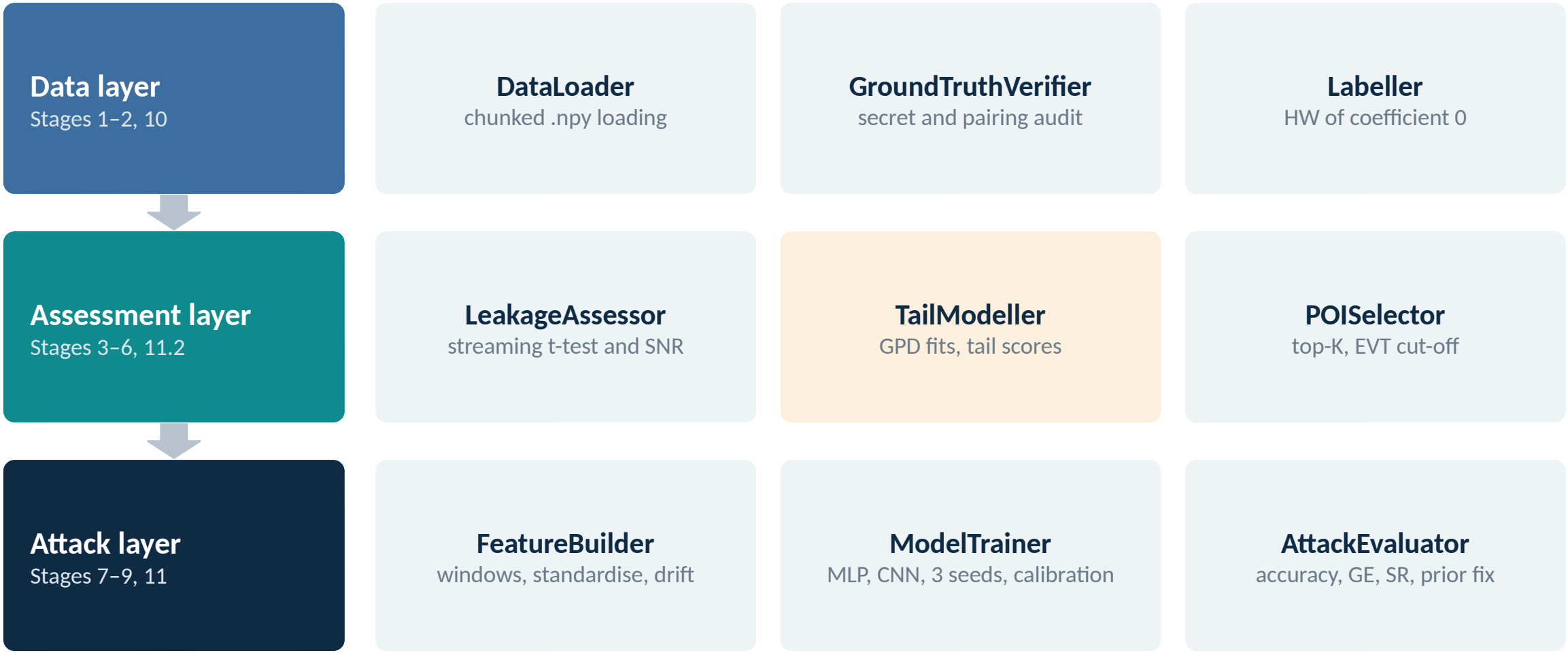
Deep-learning attack

MLP and CNN trained on profiling traces output $q(c \mid x)$; n attack traces are combined by summing log-probabilities.

Prior-aware evaluation

Accuracy vs the 23.6% baseline, guessing entropy, success rate, prior correction $\log q - \log \pi$, temperature calibration.

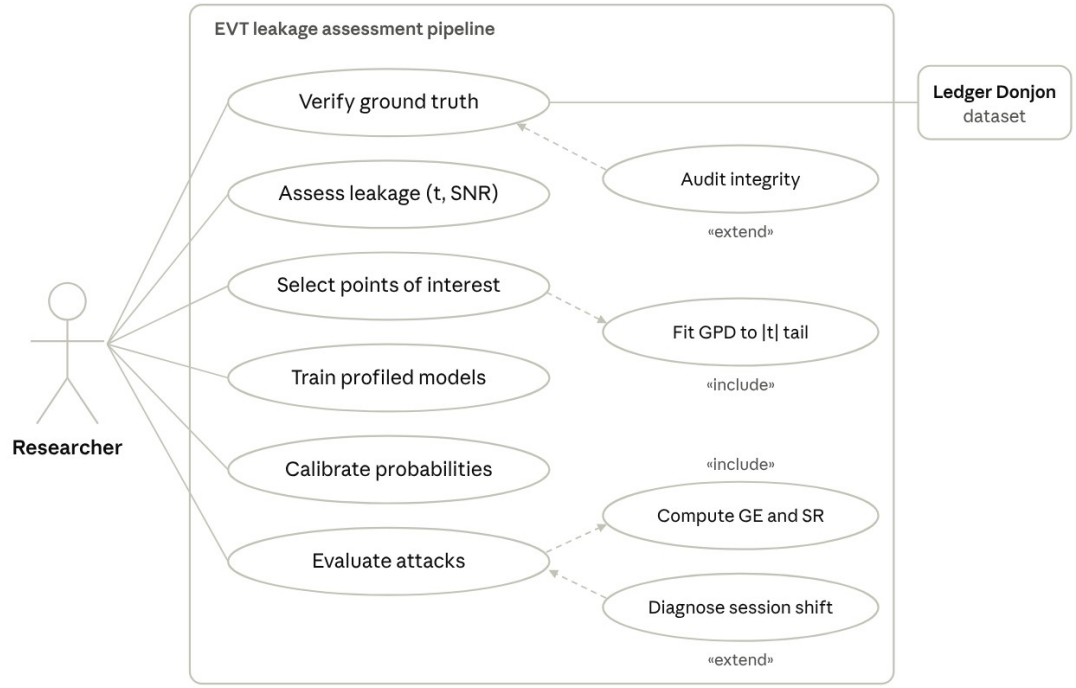
System architecture: three layers, nine modules



Google Colab + Drive; 10,000-trace streaming chunks for 8 GB sets; NVIDIA L4 GPU for Stage 11. Highlighted: EVT component.

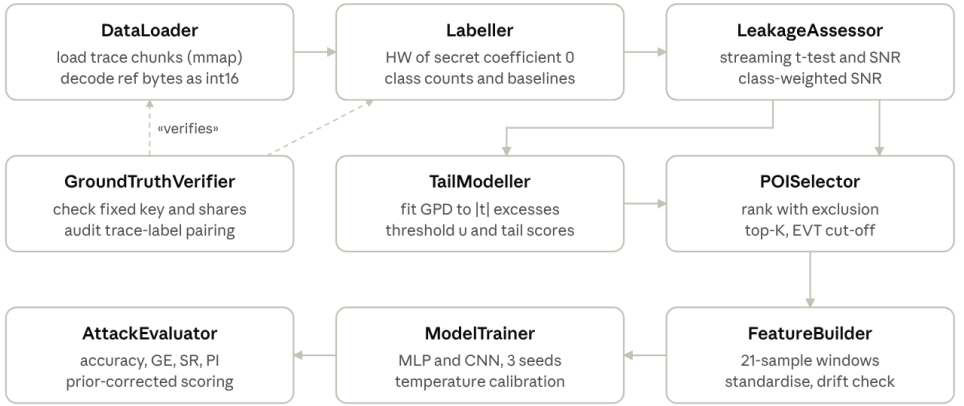
UML design: use cases and components

The researcher drives six main use cases on the dataset



Use-case diagram (report Fig. 4.3): the researcher drives six use cases; GPD fitting is included in point selection, the integrity audit extends ground-truth checks

Nine modules pass data from raw traces to attack results



Component diagram (report Fig. 4.4): nine modules pass data from raw traces to attack results

- GroundTruthVerifier checks the loader and labeller
- TailModeller (EVT) feeds POISelector
- FeatureBuilder → ModelTrainer → AttackEvaluator

Project timeline: 11 stages, three reviews

Review 1
July 2026

- 1 Dataset setup
- 2 Ground truth

Review 2
7 Oct 2026

- 3 Baseline leakage
- 4–6 POIs, GPD cut-off
- 7–8 CNN training

Final review
28 Oct 2026

- 9 Attack evaluation
- 10 Integrity audit
- 11 Max-accuracy attack

Future work
beyond this report

- C1 calibrated threshold
- C2 tail tests
- C3 EVT on evidence
- C4 generalisation

Dissertation period: 6 July 2026 to 3 November 2026. All stages 1–11 completed.

Stages 1–2 and 10: ground truth and integrity audit

Secret operand found

In the fixed-key set mult_a has 1 unique row in 200 while mult_b varies, so a is the secret; coefficient 0 explains the strongest leakage.

ref byte encoding

ref metadata is raw bytes; read as little-endian int16, 100% of values lie in [0, q) versus 5.5% big-endian.

Masked shares checked

b0 + b1 mod q equals the stored unmasked key; each share has its own 10,000-sample window.

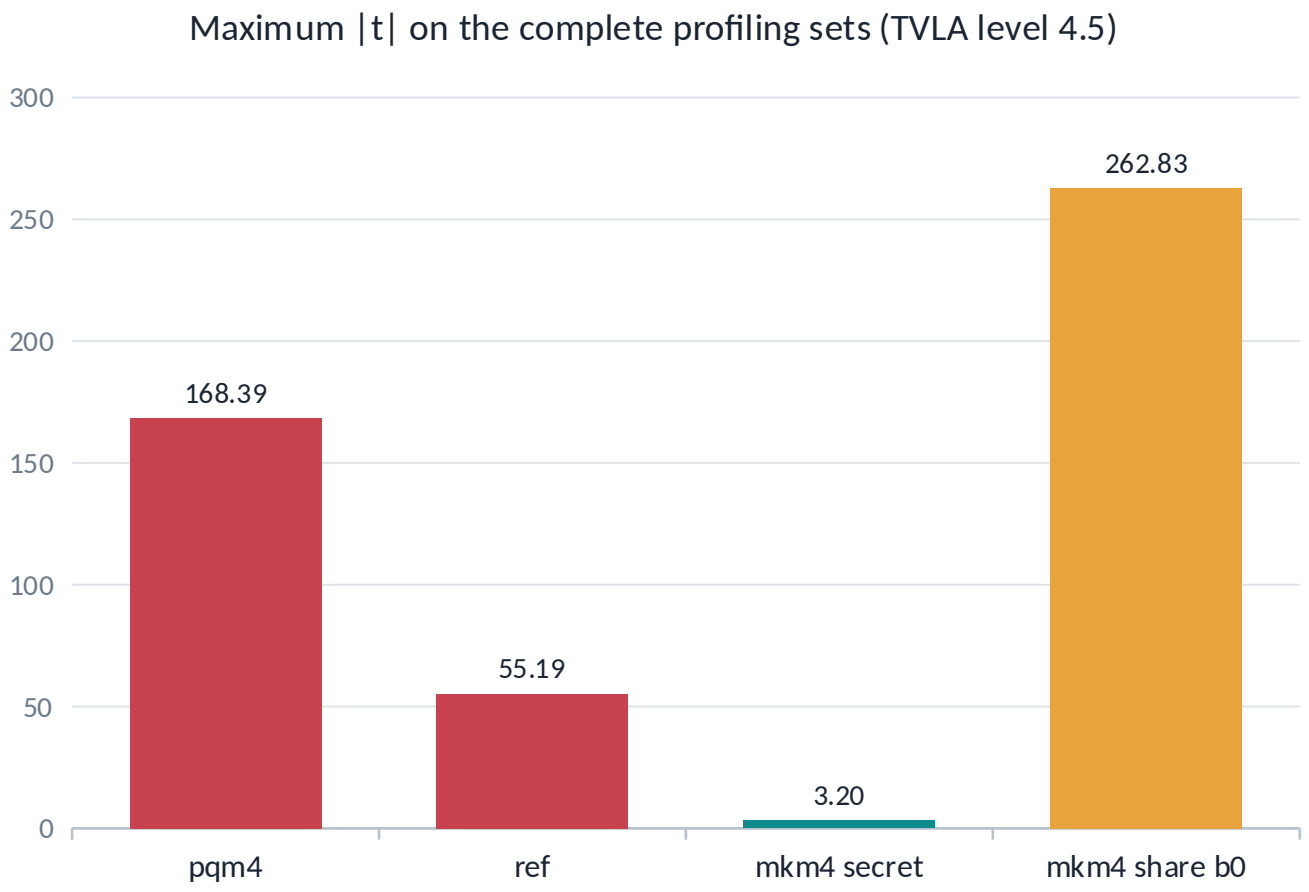
Stage 10 audit

Masked labels were misaligned with traces; rebuilt from paired share files and Stages 4–9 recomputed. pqm4 and ref passed unchanged.

Fixed secret coefficient 0

pqm4	679	HW 6 (majority class)
ref	1026	HW 2 (prior 1.98%)
mkm4	1681	HW 5

Stage 3: unprotected code leaks, masking holds



168.39 pqm4 max |t| at sample 183, SNR 1.1352

55.19 ref max |t| at sample 7917, SNR 0.2739

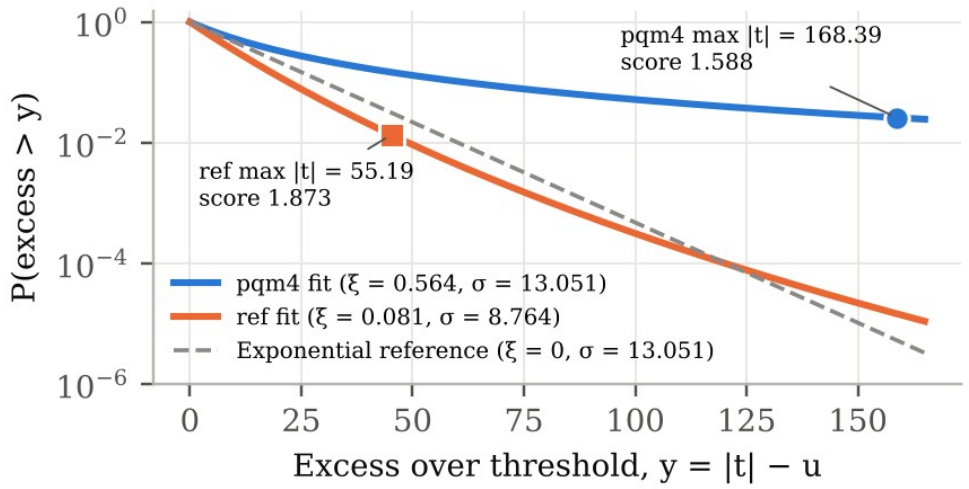
3.20 masked secret: below 4.5, no first-order leakage

262.83 positive control: share b0 alone leaks strongly

Stages 4–6: GPD tail cut-off for points of interest

GPD fit (tail 0.02)	pqm4	ref
Threshold u (98th pct of |t|)	9.676	9.964
Shape ξ / scale σ	0.564 / 13.051	0.081 / 8.764
Samples above u	400 of 20,000	260 of 13,000
Max EVT score	1.588	1.873
Points: top-50 → EVT	50 → 27 (−46%)	50 → 15 (−70%)

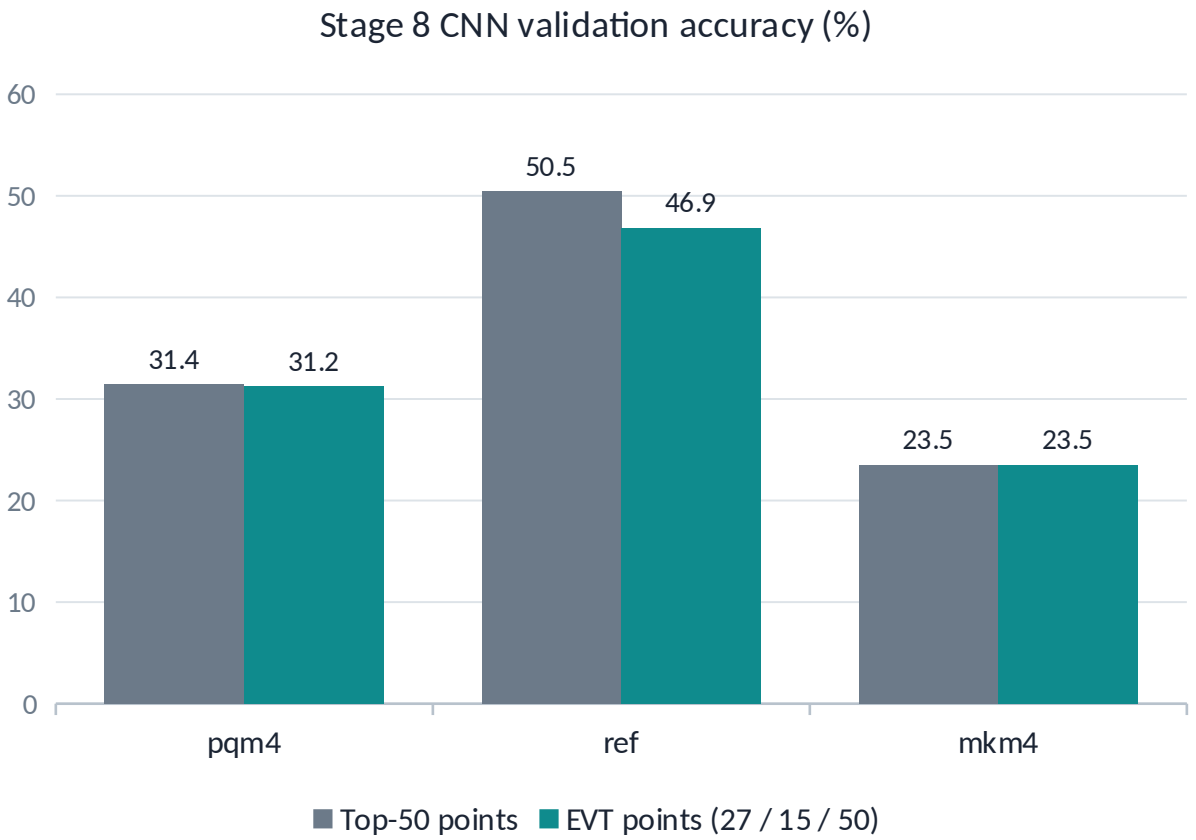
- The score keeps the |t| ranking, so only u decides the stopping point.
- The 20-sample exclusion lets each leakage peak give one point, hence 27 and 15.
- mkm4: no separation from the bulk, so 50 → 50 (0% cut).
- ξ is unstable across tail fractions (pqm4 0.59–1.16), so it is reported, not used as evidence.



Fitted GPD survival curves of the |t| tail (report Fig. 6.1)

Worked example (pqm4): y = 168.39 − 9.676 = 158.71; 1 + ξy/σ = 7.859; survival 0.0259; score $-\log_{10}(0.0259) = 1.588$.

Stages 7–9: CNN arms and first attack evaluation



Same 4,716-parameter CNN for both arms; mkm4 values are about 23.5%, the majority-class rate.

Null-corrected advantage Δ on the first 10,000 attack traces

ref top-50 / EVT
real information about HW 2

+7.3576 / +6.3712

pqm4 top-50 / EVT
moves toward HW 5: success came from the prior

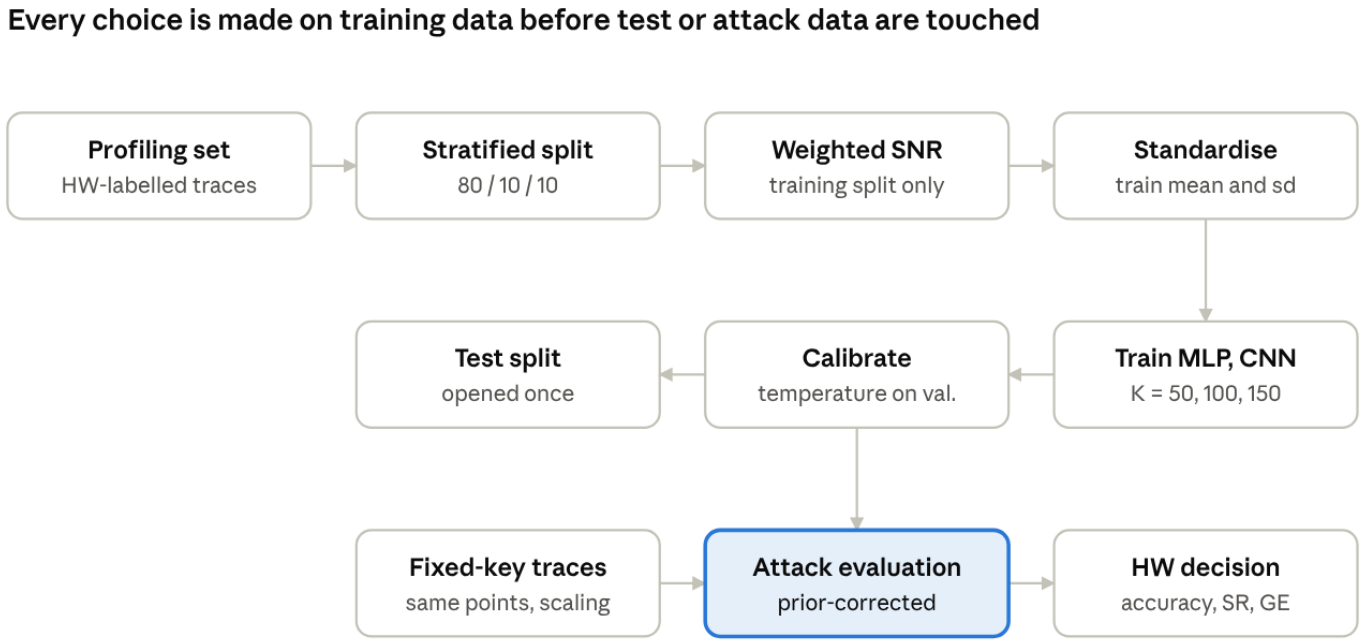
−0.4534 / −0.5068

mkm4 top-50 / EVT
noise level, as expected

−0.0197 / +0.0155

Guessing entropy saturated for pqm4 because its secret is the majority class, so a null model was needed.

Stage 11: leakage-free maximum-accuracy protocol



Stage 11 data flow (report Fig. 4.2): every choice is made on training data before test or attack data are touched

80/10/10 split · K = 50, 100, 150 · 3 seeds · temperature calibration · prior-corrected scoring

MLP

Flatten K × 21 → Dense 256 → 128 → 64 → 12 logits; batch norm, ReLU, dropout 0.3

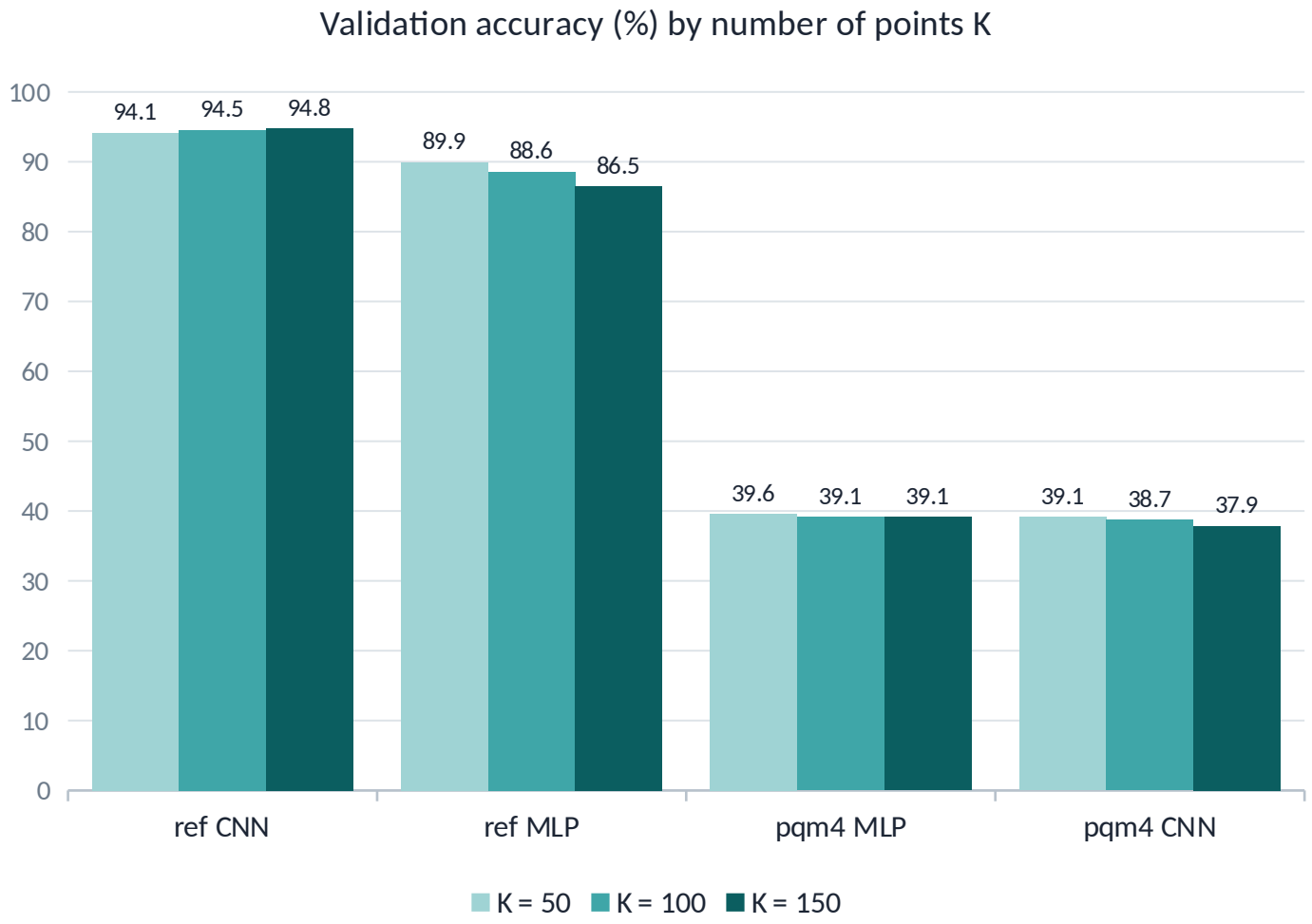
312,524 parameters at K = 50

CNN

Same 1-D filters on every 21-sample window (32, 64, 128) → Dense 256 → 128 → 12 logits

499,980 parameters; conv part only 52,736

Validation accuracy: 50 points are enough



94.1–94.8%

ref CNN: four times the 23.37% baseline; seed spread < 0.5 points

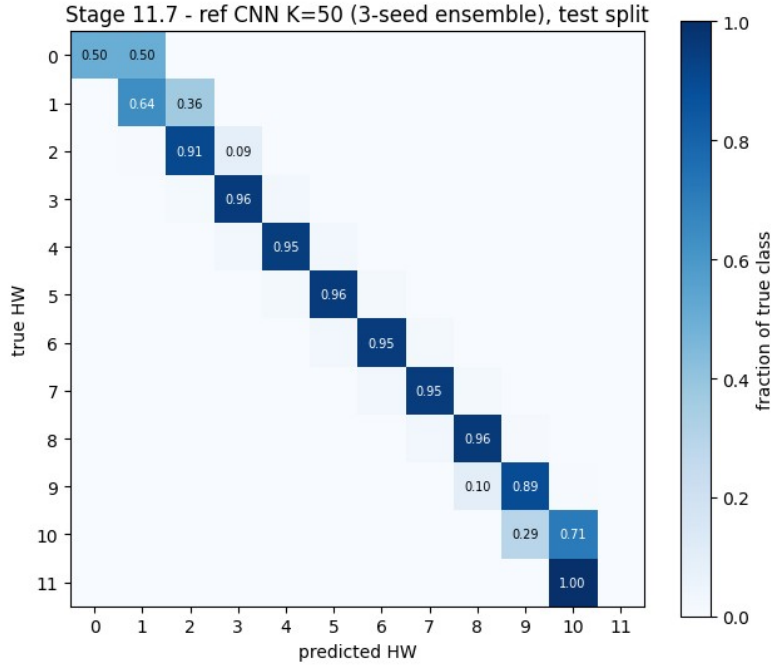
39.1–39.6%

pqm4: 1.68× the 23.61% baseline; extra points add noise

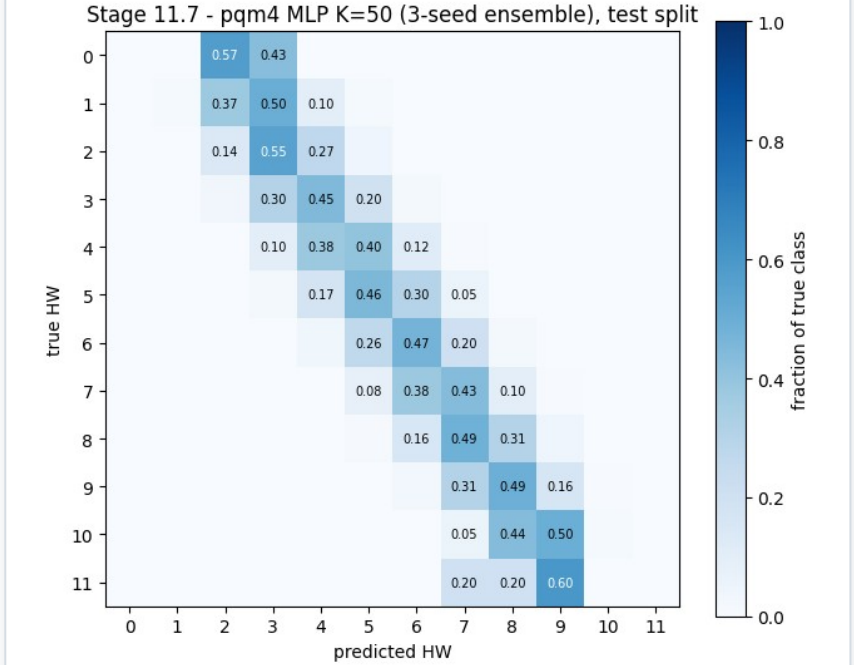
+43.7 pts

ref vs Stage 8 (50.45% → 94.13%); pqm4 +8.2 (31.42% → 39.61%)

Untouched test split confirms validation



ref CNN, K = 50 (3-seed ensemble)



pqm4 MLP, K = 50 (3-seed ensemble)

90.5%

ref attacked class HW 2 (K = 50); HW 2–9 at 89–96%

47.4%

pqm4 attacked class HW 6; HW 5–7 at 43–47%

Errors fall on neighbouring classes. pqm4 predictions are pulled toward the common middle classes, as expected with weak evidence and an unequal prior. HW 0, 1, 10 and 11 have too few test traces (as few as 2) to interpret.

Full attack sets: ref succeeds, pqm4 fails

ref CNN, K = 50 (100,000 traces)

94.65%
attack accuracy

0.892
success, 1 trace

1
trace to GE < 0.5

Decision HW 2 = true class; 10 traces always succeed.

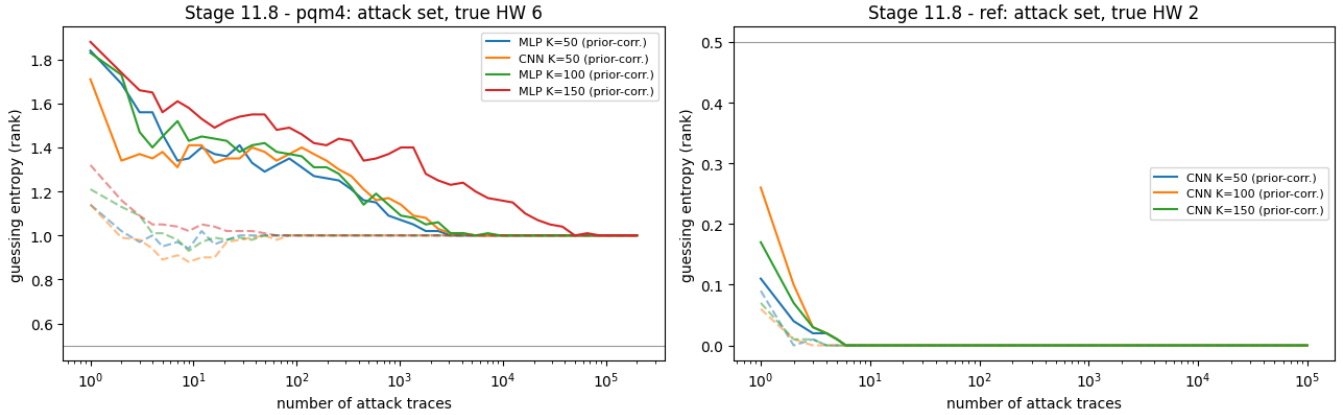
pqm4 MLP, K = 50 (200,000 traces)

32.38%
attack accuracy

0.237
success, 1 trace

0
success, 100 traces

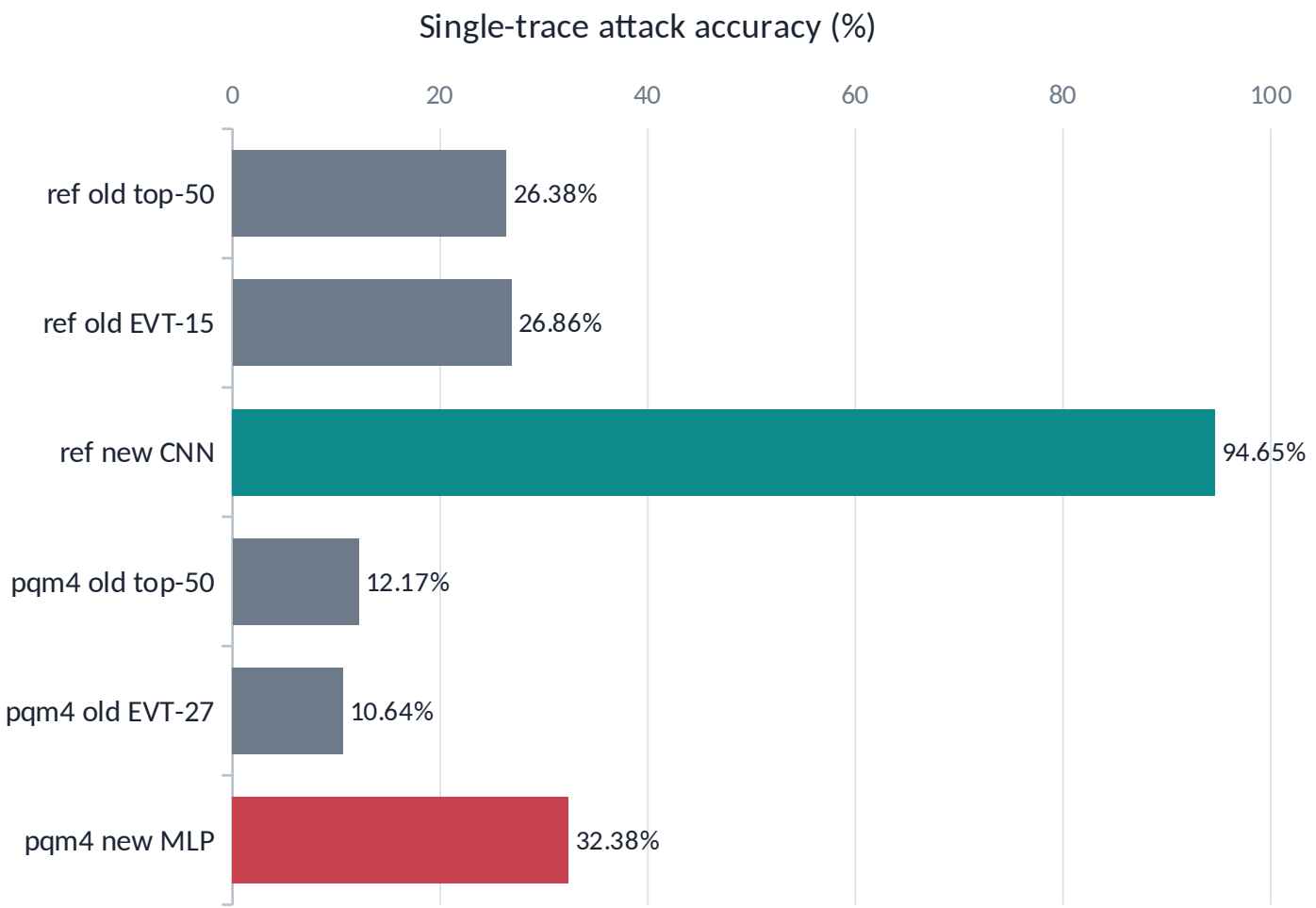
Every chunk and model decides HW 5; true HW 6 ranks second.



Guessing entropy on the full attack sets (report Fig. 7.4)

A success rate that falls as traces are added means the pqm4 attack is consistently biased toward a wrong answer, not merely weak.

Stage 8 versus Stage 11 on the full attack sets



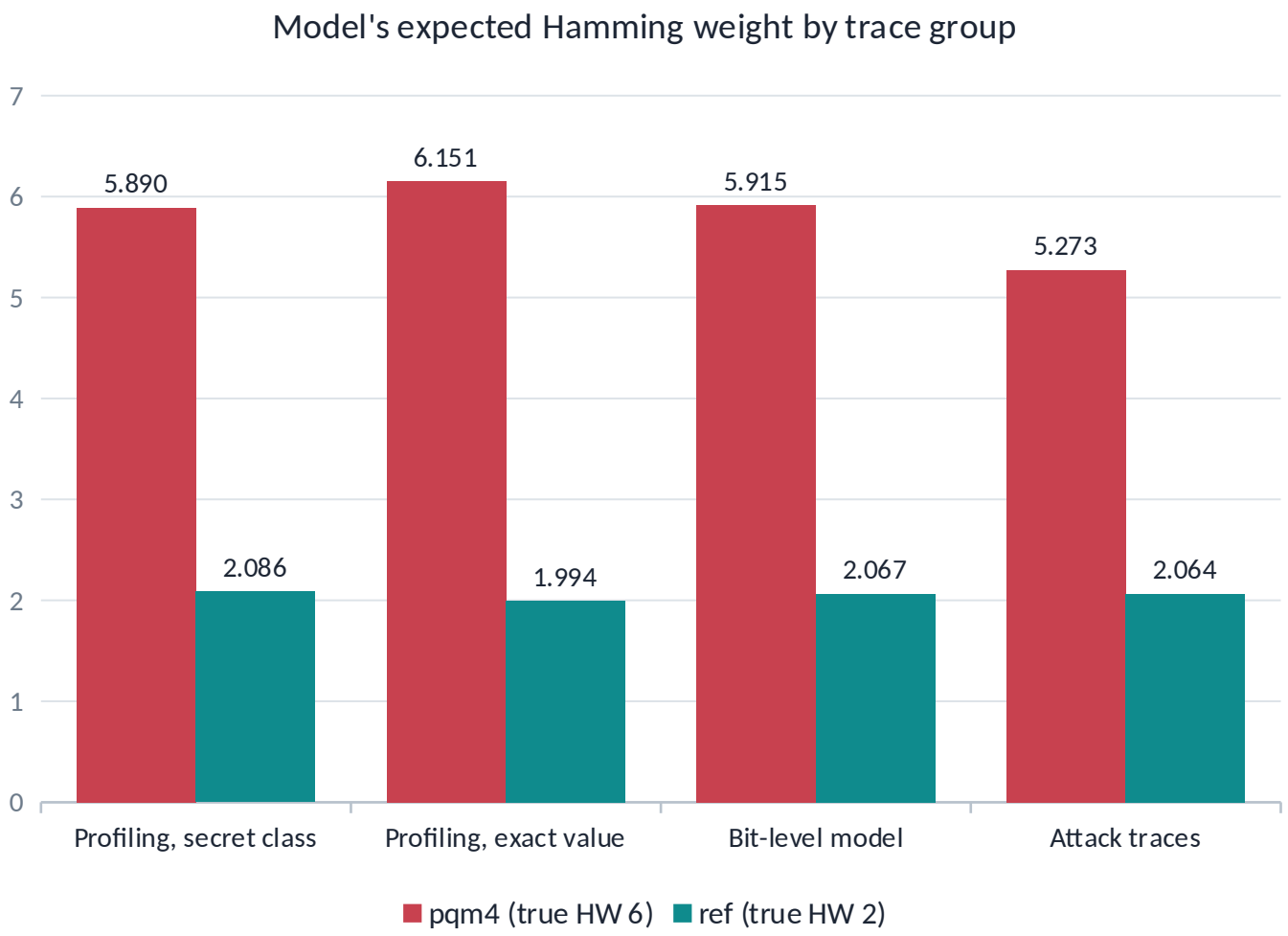
Large, real ref gain
≈26% → 94.7%; traces for GE < 0.5 fall from 4 to 1.

Prior correction rescued old ref
Plain scoring picks HW 3; corrected scoring finds HW 2 within 4 traces.

EVT cut-off cost nothing on ref
15 points need the same 4 traces; SR at 10 traces 0.909 vs 0.897.

pqm4 closer, not solved
True class rises from rank 2–6 to rank 1; decision still HW 5.

Why pqm4 fails: a recording-session shift



Not the model

Profiling traces of value 679 are read as about HW 6 (5.89, 6.15, 5.92).

Not the value

Bit model places 679 at the 57th percentile of difficulty (R^2 0.683).

The session

Attack traces read as HW 5.27; attack drift 0.120 vs 0.0058 on validation (21×), equal in leaky and noise windows, spread unchanged (0.98).

Conclusion: answers to the research questions

RQ1 pqm4 and ref leak strongly (peaks at samples 183 and 7917); the masked implementation does not leak the secret at first order.

RQ2 GPD cut-off keeps 27 and 15 points instead of 50, costing 0.2 and 3.6 points of validation accuracy; the stopping point depends on the tail fraction.

RQ3 ref CNN 94.1–94.8% and pqm4 MLP 39.1–39.6% validation, little change from K = 50 to 150; ref holds on test and attack sets, pqm4 on the test split only.

RQ4 The pqm4 failure is a shift between profiling and attack recording sessions, not the model or the secret value.

Contributions

- Audited assessment of all three implementations, with the dataset corrections it needs
- Empirical test of a GPD tail cut-off: 46–70% fewer points
- Leakage-free attack: 94.65% single-trace accuracy, HW class from one trace (ref)
- Diagnosed cross-session failure (pqm4)
- Prior effects demonstrated on an ML-KEM dataset

Limitations and future work

Limitations

- One coefficient (0) and one leakage model (HW); a class is recovered, not a key
- One secret per attack set; pqm4 diagnosis rests on 12 exact-value traces
- pqm4 session shift diagnosed, not corrected
- Unequal seeds for a few configurations; no ablation of Stage 11 changes
- EVT cut-off depends on the tail fraction; no declustering
- Masked implementation analysed at first order only

Future work

C1	Calibrated detection threshold from the trace-wide maximum under no leakage
C2	Tail-sensitive class tests beyond the mean
C3	EVT applied to per-trace attack evidence
C4	All 256 coefficients and a second public dataset
+	Domain adaptation for pqm4; second-order attack on mkm4

Key references

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Full list of 20 references (APA 7) in Chapter 9 of the report.

Thank You

Questions and discussion

G Madhusudan | 25MCY0001 | Guide: Dr. Jeyanthi N